CM2108 Secure Communication Networks

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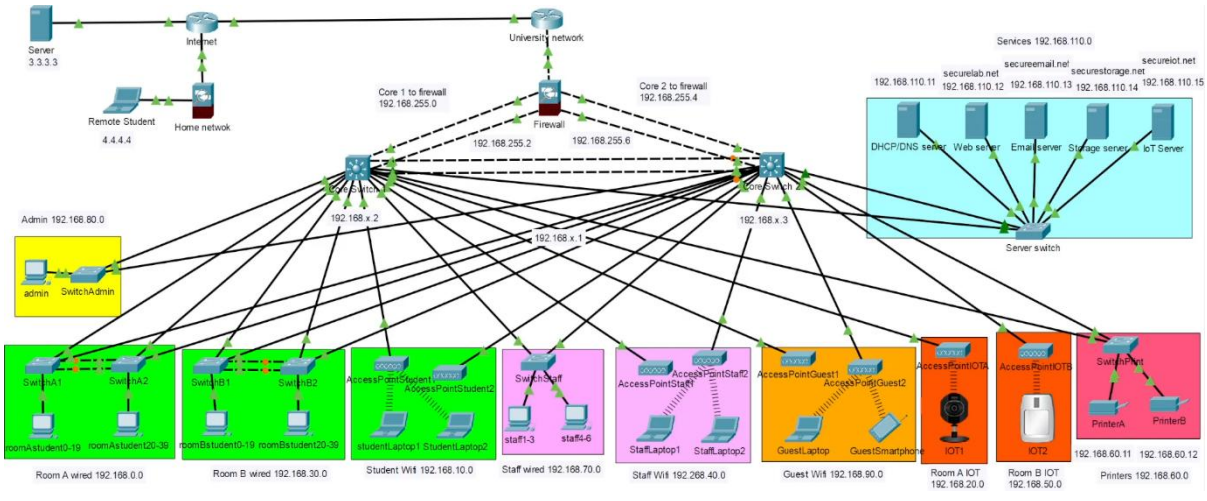
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Network design

This network design accommodates a range of devices used across two classrooms.

Assumptions I have made from the scenario:

- The network is sandboxed from the University Network, so the network must be physically isolated and all external traffic should be untrusted by default.
- The two classrooms will have 30-40 students each who each need access to a wired PC.
- There will be 1-3 staff members in each room who also need access to a wired PC.
- Staff and students will have wireless devices which will also need network access.
- Students and staff will need shared access to printers and IoT devices.
- The lab needs its own local services separate from the University.
- Guest access is required.



The network is segmented to provide separate access levels for students, staff, guests, admin, IoT devices, printers and servers. Student and staff networks are further separated by connection type and additionally students and IoT into room A and B to minimise broadcast domains.

The admin PC, along with all servers and network devices, will be kept in a third secure room. This room will be kept locked and restricted from all except administrators.

Services that the lab offers include:

- HTTPS – Useful for demonstrations, students could use for testing.
- DHCP/DNS – Assigns addresses for the network and gives a domain name to the web server.
- SMTP/POP3 – Email is essential for any modern school environment.
- FTP – Students and staff can store files locally.
- IoT – Useful for both security and teaching purposes.

Using routers to segment this network would require a large web of devices and cabling, resulting in a high costing and difficult to maintain network. Taking inspiration from Cisco's Collapsed Core design, I have decided to use a layer of redundant multilayer switches instead of routers to route traffic between subnets. This makes the network more accessible, scalable and efficient.

The gateway to the network is guarded by an ASA firewall which monitors and restricts inbound traffic.

Following guidance from NetworkLessons.com, I have configured an IPsec site-to-site VPN between the two ASAs to simulate secure remote access. However, I didn't manage to get the encryption to function in practice.

Subnetting scheme

Description	End devices	Network Id	Subnet mask	Usable hosts	DHCP pool	VLAN	Access group
Room A Wired	40	192.168.0.0/26	255.255.255.192	62	192.168.0.11 – 192.168.0.50	10	student
Student WiFi	100	192.168.10.0/25	255.255.255.128	126	192.168.10.11 – 192.168.10.110	11	student
Room A IoT	100	192.168.20.0/25	255.255.255.128	126	192.168.20.11 – 192.168.20.110	12	IoT
Room B Wired	40	192.168.30.0/26	255.255.255.192	62	192.168.30.11 – 192.168.30.50	13	student
Staff WiFi	100	192.168.40.0/25	255.255.255.128	126	192.168.40.11 – 192.168.40.110	14	staff
Room B IoT	100	192.168.50.0/25	255.255.255.128	126	192.168.50.11 – 192.168.50.110	15	IoT
Printers	2+	192.168.60.0/27	255.255.255.224	30	Static	16	printer
Staff wired	6+	192.168.70.0/27	255.255.255.224	30	Static	17	staff

Admin	1+	192.168.80.0/27	255.255.255.224	30	Static	18	admin
Guest WiFi	100	192.168.90.0/25	255.255.255.128	126	192.168.90.11 – 192.168.90.110	19	guest
Services	4+	192.168.110.0/27	255.255.255.224	30	Static	21	server
Core 1 to firewall	2	192.168.255.0/30	255.255.255.252	2	Static	100	internal
Core 2 to firewall	2	192.168.255.4/30	255.255.255.252	2	Static	101	internal

Subnets have been chosen to separate traffic by trust level.

- Students and staff need access to all the lab's services, the internet, printers and IoT devices.
- Staff also need to be able to communicate with student and guest devices for supervision.
- IoT, printers and servers should be restricted from initiating communication with other subnets and should only be able to respond using essential services only.
- Guests should be restricted to web/DHCP services and internet only.
- Admins need unrestricted access across the network, and all subnets must be able to communicate to the admin subnet.

Addresses are given a consistent scheme to assist management.

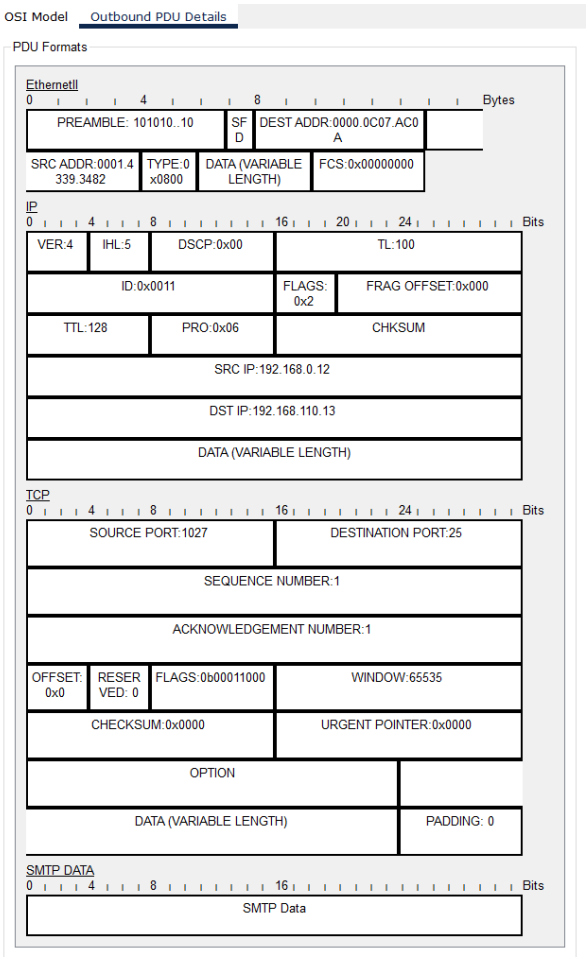
- Plenty of space has been left between each subnet for future growth.
- .1 is always the default gateway for each subnet
- .2 - .10 are reserved for network devices, as recommended by LearnTech Training on YouTube
- All end device addresses start from .11

Subnets are chosen to fit end devices comfortably while allowing for growth.

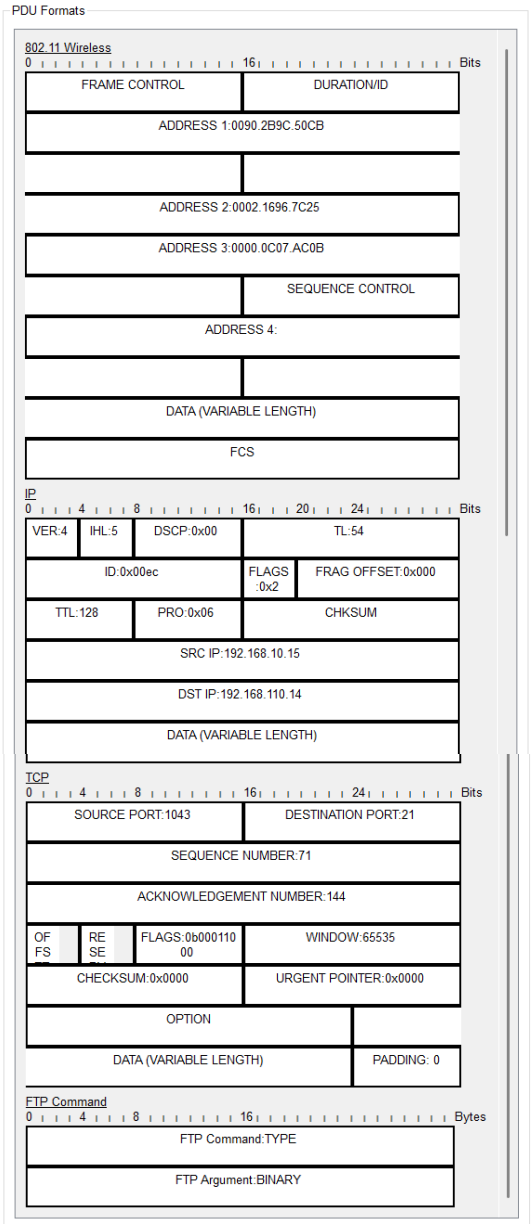
- Student wired subnets are given /26 networks to comfortably fit 40 PCs per room to accommodate large classes.
- All Wi-Fi subnets are given /25 networks to fit 100 devices.
    - Users will likely have 2+ wireless devices.
    - On some occasions a higher number of guests or staff may need access to the network, such as during open days or conferences.
- Smaller subnets are given a larger-than-needed /27 network.
    - While there is only one device in the admin network now, admins may want to set up additional devices for security and monitoring.
    - Similarly, staff members may need more wired connections for additional devices.
    - Other devices such as backup servers may be added to the services subnet.
- Point to point subnets only need 2 addresses so are assigned /30.

Networking services and protocols

Suppose a student on a wired PC in room A presses send on an email.



- 7-Application layer – Student's email application will use SMTP to communicate with the lab's email server and wraps the message data accordingly.
- 6-Presentation layer – Encoded using the MIME protocol to preserve uncommon data types such as foreign characters. All SMTP data is encrypted using TLS.
- 5-Session layer – A TLS session is established.
- 4-Transport layer – TCP breaks the data into segments, adding headers:
    - Source port – a random port on the student PC.
    - Destination port – SMTP uses port 25.
    - Sequence number – Ensures data is reassembled correctly.
    - Acknowledgement number – Confirms that data has been received.
    - Checksum – Detects missing data, which is to be retransmitted.
- 3-Network layer – IP encapsulates each segment with:
    - Source IP – Student PC.
    - Destination IP – Email server.
    - A time-to-live.
    - Checksum – Detects IP corruption.
- 2-Data link layer – Ethernet adds the following headers:
    - Source MAC – Student PC.
    - Destination MAC – The PC checks its ARP table and adds the MAC of the default gateway.
- 1-Physical layer – Packets are converted to electric signals and leave through the NIC onto copper cables.

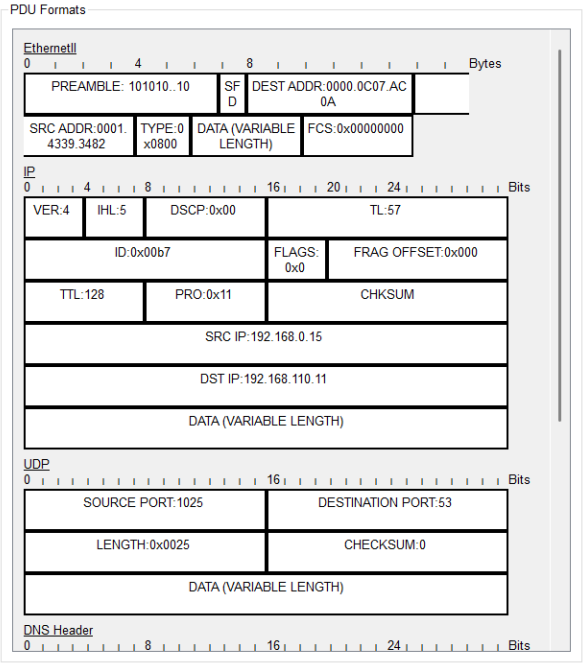


Now suppose a student connected by Wi-Fi uploads a file to the storage server remotely.

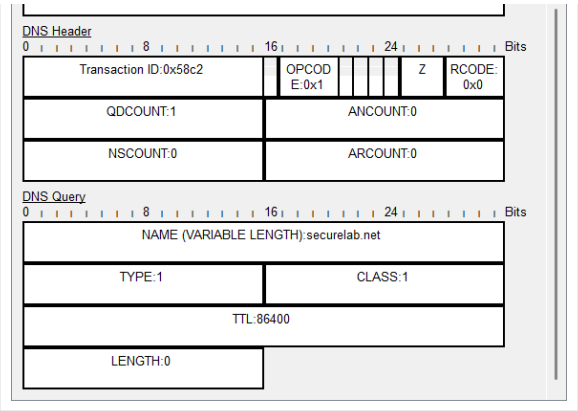
- 7 – FTP creates metadata for the file such as a filename. It uses a control channel and a data channel.
- 6 – Data may be compressed for faster transfer.
- 5 – Two separate sessions are kept for control and data.
- 4 – TCP segments data and adds headers:
  - Destination port – 21 for control, random for data.
  - Source port is random.
  - Sequence numbers and checksums.
- 3 – IPs of the student device and server are added, along with a TTL and header checksum.
- 2 – Wi-Fi adds the headers:
  - MAC 1 – Student device.
  - MAC 2 – The Wireless Access Point.
  - MAC 3 – The identifier of the wireless network.
- 1 – NIC converts data to radio waves and transmits them through the air. The WAP receives them and converts them back into electric signals, before forwarding to the core.

Now suppose another student on a wired PC types ‘securelab.net’ into their browser.

- 7 – The browser generates a DNS query. Once the web server’s IP is known it sends an HTTPS request.
- 6 – HTTPS uses TLS to encrypt all traffic.
- 5 – A HTTPS session is established once the DNS query is resolved.
- 4 – DNS uses UDP, whereas HTTPS uses TCP. Both add headers:
  - Random source port.
  - Destination port of 53 for DNS, 443 for HTTPS.
  - Checksum.
  - Only TCP uses a sequence and acknowledgement number.
- 3 – IP adds the headers:
  - Source IP – Student PC.
  - Destination IP – First the DNS server, then the web server for HTTPS.
  - TTL and checksum.
- 2 – Ethernet adds:
  - Source MAC – Student PC



    - Destination MAC – Default gateway from PC's ARP table.
- 1 – Data is sent out the NIC onto copper cables.



Routing/switching

- When receiving packets from end devices, the access switch or WAP checks its MAC table and forwards them via Ethernet to the core.
- Packets sent to the standby core switch are dropped by HSRP.
- The active core switch decapsulates packets to layer 3, decrements TTL and recognises the destination IP as belonging to the services VLAN.
- After checking its ARP table, the core re-encapsulates packets with the respective server's MAC.
- Packets are forwarded to the services access switch, before finally reaching the server where they are decapsulated and the application layer data is processed.
- The same process is applied on return of packets.

Network Security and Availability (A)

One primary security mechanism implemented is the use of Access Control Lists on the core switches and ASA firewall. ACLs restrict traffic by analysing packets at layer 3 and 4, ensuring only authorised devices can access specific services. I have given hosts only the minimum permissions they require, upholding the least privileges principle.

On my core ACL for example I have configured the services ACL to only permit the return traffic needed to respond to users, and an explicit deny at the end blocks all other traffic. On the ASA firewall, access into the network is restricted to web and email servers only using their required ports. This reduces the risk of network scanning, brute force attempts and malware delivery. Confidentiality is upheld by preventing unauthorised access.

To help uphold integrity I have implemented DHCP snooping, as recommended by CCNA Practical Labs. DHCP snooping prevents untrusted servers from assigning addresses to devices. This helps to prevent man-in-the-middle attacks where an attacker pretends to be a DHCP server and intercepts traffic. Only the port connecting my trusted DHCP server is allowed to send DHCP responses. Additionally, I have helped to ensure availability by applying rate-limits to each of the core's SVIs to prevent DHCP starvation attacks.

DHCP snooping automatically builds a list of trusted IP-MAC bindings, enabling me to activate Dynamic ARP Inspection. DAI inspects traffic and blocks any ARP responses that do not match these trusted bindings, further upholding integrity by preventing ARP poisoning attacks.

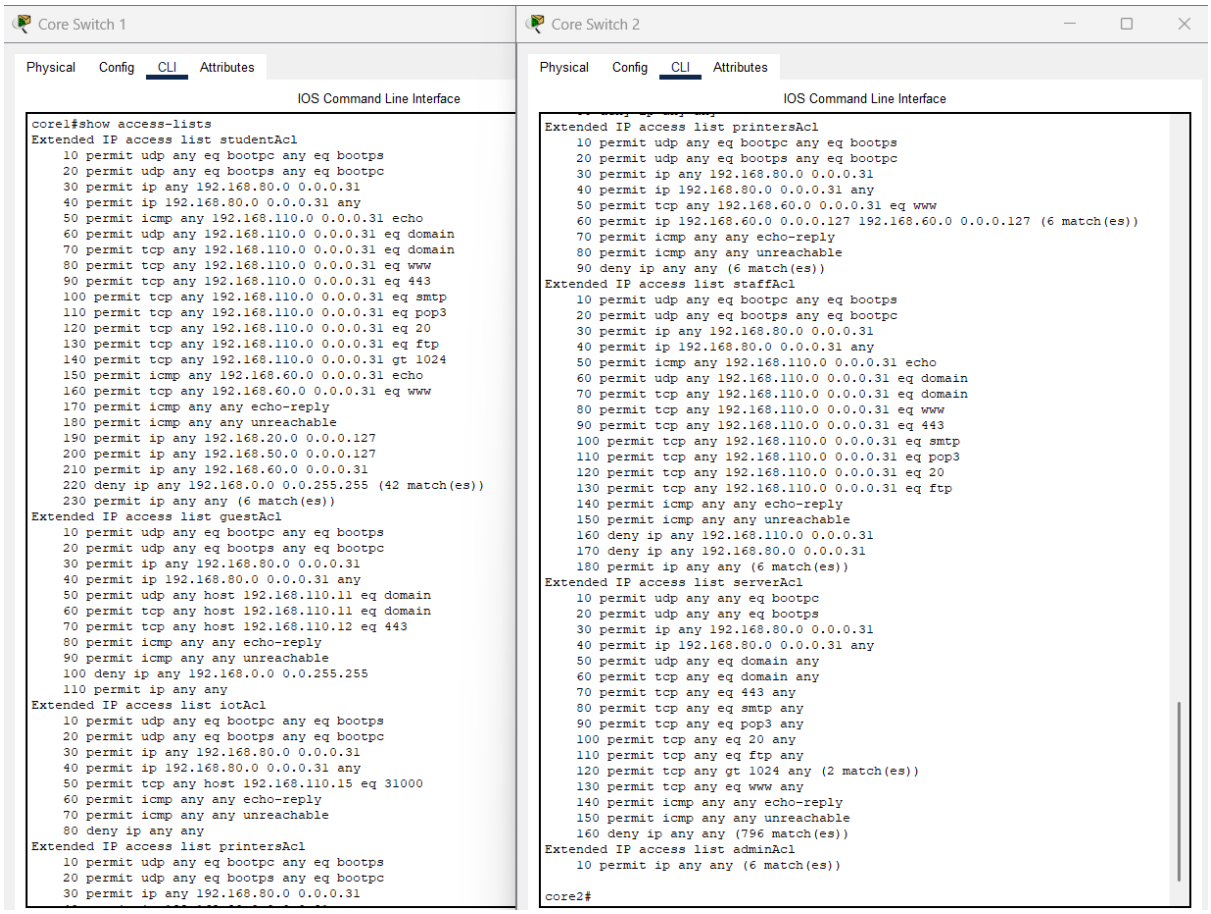
To simulate secure remote access, I have attempted to implement an IPsec site-to-site VPN to a remote student's network. This would encrypt all traffic and authenticate both endpoints, insuring confidentiality and integrity when data is passing through untrusted networks.

Admin access to network devices is also secure, devices are password protected and would have physical management ports blocked off. Management is permitted only to the admin network via SSH. The admin room will be kept locked with key-card access. 24/7 CCTV will run internally and at all entrances to the lab. These measures help to ensure confidentiality by preventing tampering.

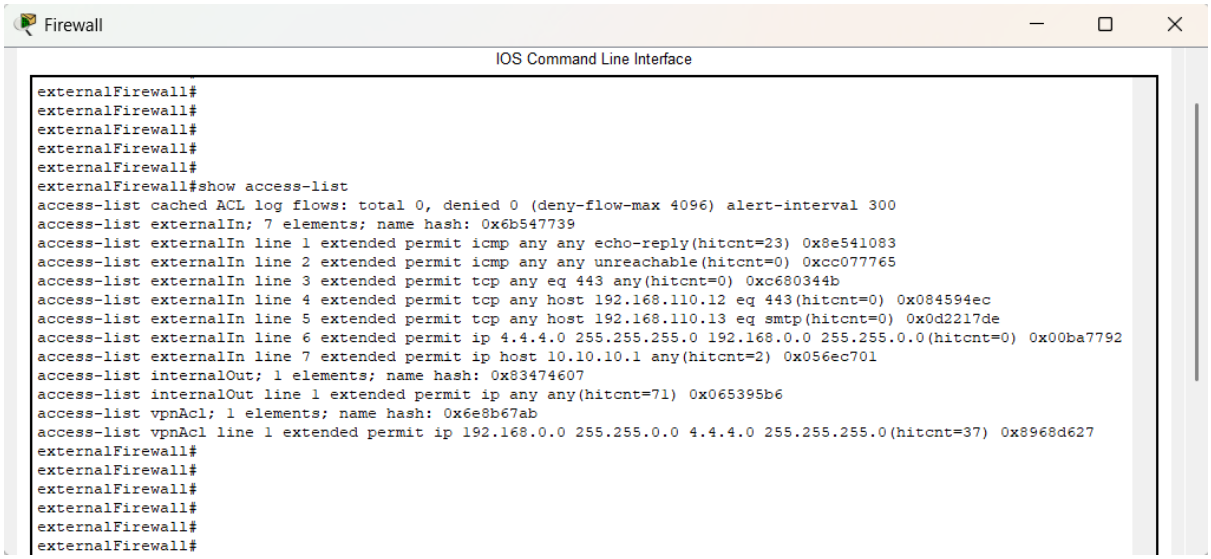
To further uphold the confidentiality and integrity of my network I should implement a central Authentication, Authorization and Accounting server. This would ensure only authenticated users can access the network and provide a log of admin activity. To avoid compromising availability, a backup AAA server would be needed to prevent a single point of failure.

Configurations

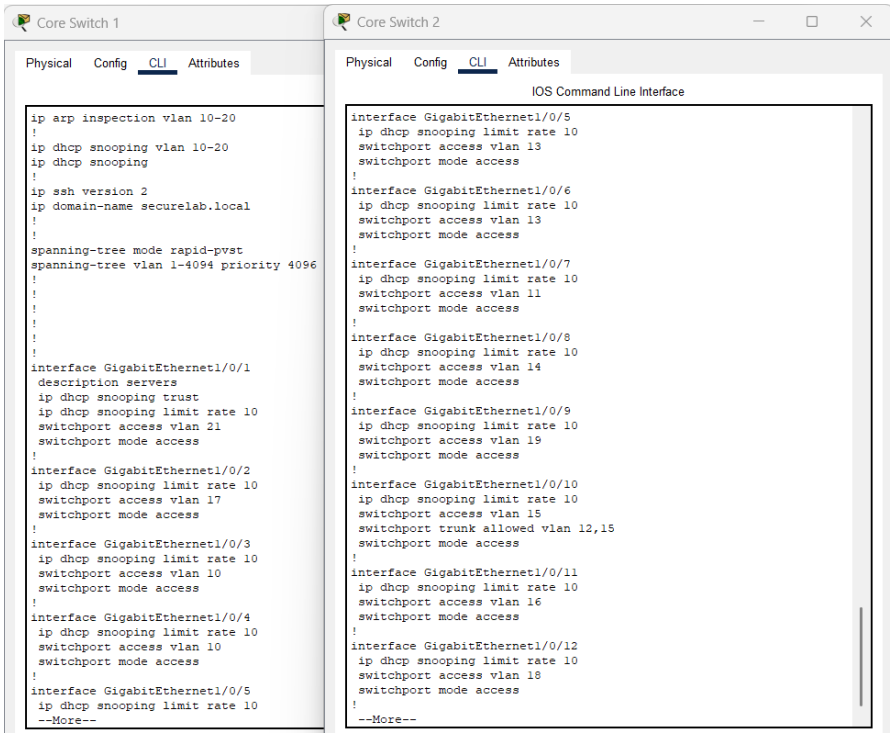
Core switch ACL



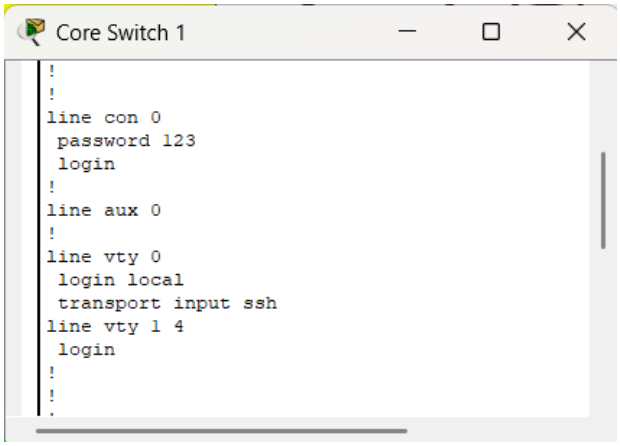
Firewall ACL



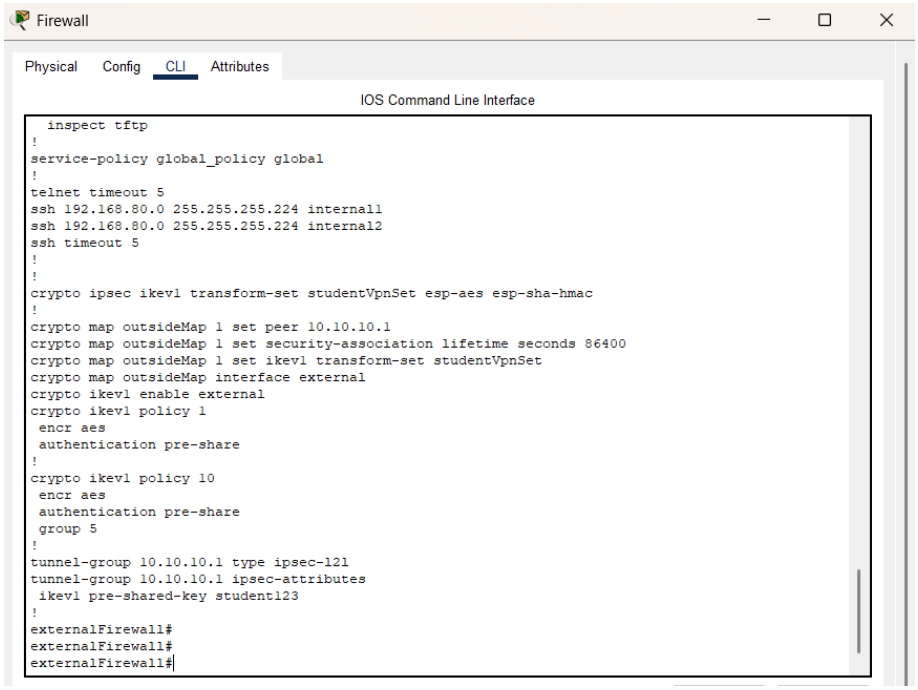
DHCP snooping and ARP inspection



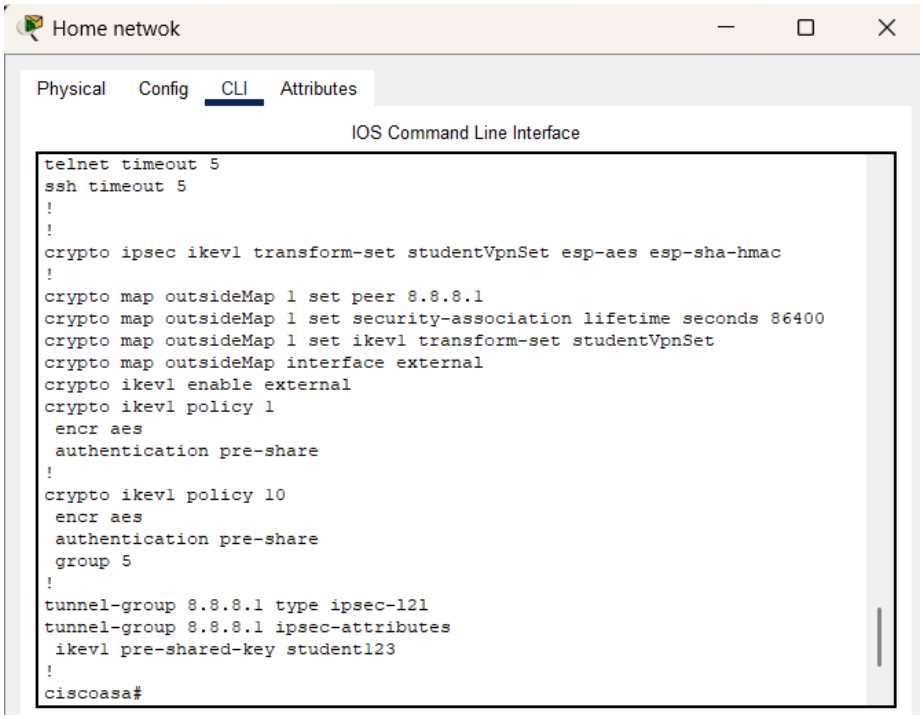
Core switch password protection and SSH access



Firewall ssh access and IPsec encryption:



Remote student's IPsec encryption



Network Security and Availability (B)

The core layer of my topology consists of two redundant multilayer switches.

- Configured with HSRP to provide a virtual interface shared between both devices.
- For each VLAN, one switch acts as active and one remains standby.
- Each access switch connects to both core switches.
- If one switch fails, the standby is promoted to active. Devices using the virtual interface do not need to change configuration.
- Constant HSRP 'hello' messages keep failover time to a few seconds.

Similarly, my ASA firewall uses EtherChannel to bundle internal interfaces into single logical links. This provides redundancy whilst also increasing bandwidth. If a single or link has a malfunction, HSRP/EtherChannel automatically keeps the network running. This prevents single points of failure and ensures availability even during faults.

High performance is ensured by using hardware level inter-VLAN routing, providing lower latency and higher throughput compared to software-based routing.

Network performance and availability can be measured using metrics such as:

- Failover time – How quickly redundancy methods take effect.
- Packet loss/latency – Indicates congestion or link failures.
- Throughput – Shows how efficiently bandwidth is being used.
- Uptime – Shows the overall network dependability.

Configurations

Core switch SVIs shared using HSRP

Core Switch 1

Physical Config CLI Attributes

IO

```
interface Vlan10
 mac-address 0010.11d3.3e01
 ip address 192.168.0.2 255.255.255.192
 ip helper-address 192.168.110.11
 ip access-group studentAcl in
 standby 10 ip 192.168.0.1
 standby 10 priority 150
 standby 10 preempt
!
interface Vlan11
 mac-address 0010.11d3.3e02
 ip address 192.168.10.2 255.255.255.128
 ip helper-address 192.168.110.11
 ip access-group studentAcl in
 standby 11 ip 192.168.10.1
 standby 11 priority 150
 standby 11 preempt
!
interface Vlan12
 mac-address 0010.11d3.3e03
 ip address 192.168.20.2 255.255.255.128
 ip helper-address 192.168.110.11
 ip access-group iotAcl in
 standby 12 ip 192.168.20.1
 standby 12 priority 150
 standby 12 preempt
!
interface Vlan13
 mac-address 0010.11d3.3e04
 ip address 192.168.30.2 255.255.255.192
 ip helper-address 192.168.110.11
 ip access-group studentAcl in
 standby 13 ip 192.168.30.1
 standby 13 preempt
!
interface Vlan14
 mac-address 0010.11d3.3e05
 ip address 192.168.40.2 255.255.255.128
 ip helper-address 192.168.110.11
 ip access-group staffAcl in
 standby 14 ip 192.168.40.1
 standby 14 preempt
!
interface Vlan15
 mac-address 0010.11d3.3e06
 ip address 192.168.50.2 255.255.255.128
 ip helper-address 192.168.110.11
 ip access-group iotAcl in
 standby 15 ip 192.168.50.1
 --More--
```

Core Switch 2

Physical Config CLI Attributes

IOS Command

```
 standby 15 preempt
!
interface Vlan16
 mac-address 0002.1690.1d07
 ip address 192.168.60.3 255.255.255.224
 ip helper-address 192.168.110.11
 ip access-group printersAcl in
 standby 16 ip 192.168.60.1
 standby 16 preempt
!
interface Vlan17
 mac-address 0002.1690.1d08
 ip address 192.168.70.3 255.255.255.224
 ip helper-address 192.168.110.11
 ip access-group staffAcl in
 standby 17 ip 192.168.70.1
 standby 17 priority 150
 standby 17 preempt
!
interface Vlan18
 mac-address 0002.1690.1d09
 ip address 192.168.80.3 255.255.255.224
 ip helper-address 192.168.110.11
 ip access-group adminAcl in
 standby 18 ip 192.168.80.1
 standby 18 priority 150
 standby 18 preempt
!
interface Vlan19
 mac-address 0002.1690.1d0a
 ip address 192.168.90.3 255.255.255.128
 ip helper-address 192.168.110.11
 ip access-group guestAcl in
 standby 19 ip 192.168.90.1
 standby 19 preempt
!
interface Vlan20
 mac-address 0002.1690.1d0b
 ip address 192.168.100.3 255.255.255.128
 ip helper-address 192.168.110.11
 standby 20 ip 192.168.100.1
 standby 20 priority 150
 standby 20 preempt
!
interface Vlan21
 mac-address 0002.1690.1d0c
 ip address 192.168.110.3 255.255.255.224
 ip access-group serverAcl in
 standby 21 ip 192.168.110.1
 standby 21 preempt
```

EtherChannel interfaces on firewall

